Here is the complete project documentation for your foreign travel expense tracking and budget management program, structured just like the previous ones!

1. Problem Statement, Objectives, and Assumptions

Problem Statement

Managing travel expenses across multiple foreign currencies requires converting currencies, accounting for transaction fees, aggregating overall spending, identifying the single largest expenditure, and comparing the final sum against a predefined budget. Manual tracking is tedious and prone to calculation errors. An automated program is needed to streamline foreign travel cost management.

Objectives

- **Automate Currency Conversion & Fees:** To instantly convert foreign currency amounts into local currency (Rupees) using exchange rates and automatically apply a transaction fee (2%).

- **Aggregate and Track Expenses:** To compute the total travel cost and dynamically track the category with the highest expenditure.

- **Budget Validation:** To compare total accumulated expenses against a user-defined travel budget and determine financial compliance.

Assumptions

- The number of expenses and financial inputs (amounts, exchange rates, budget) are valid numerical values.
- A standard flat transaction fee of 2% applies to all converted rupee amounts.
- Budget and final totals are measured in a consistent currency unit (Rupees).

2. Algorithm, Flowchart, and Pseudocode

Algorithm

1. **Start** the program.
2. **Input** n (number of expenses) as an integer.
3. **Initialize** total = 0, largest = 0, and largest_name = "".
4. **Repeat** n times using a loop:

- **Input** expense name, foreign currency amount, and exchange rate.
- Calculate converted amount: rupees = amount * rate.
- Calculate fee: fee = rupees * 0.02.
- Calculate total item cost: cost = rupees + fee.
- Update aggregate total: total = total + cost.
- Check if cost > largest: if true, update largest = cost and largest_name = name.

5. **Display Results:** Print Total Travel Cost, Largest Expense Name, and Largest Expense Cost.

6. **Input** budget as a floating-point number.

7. **Evaluate Budget Condition:**

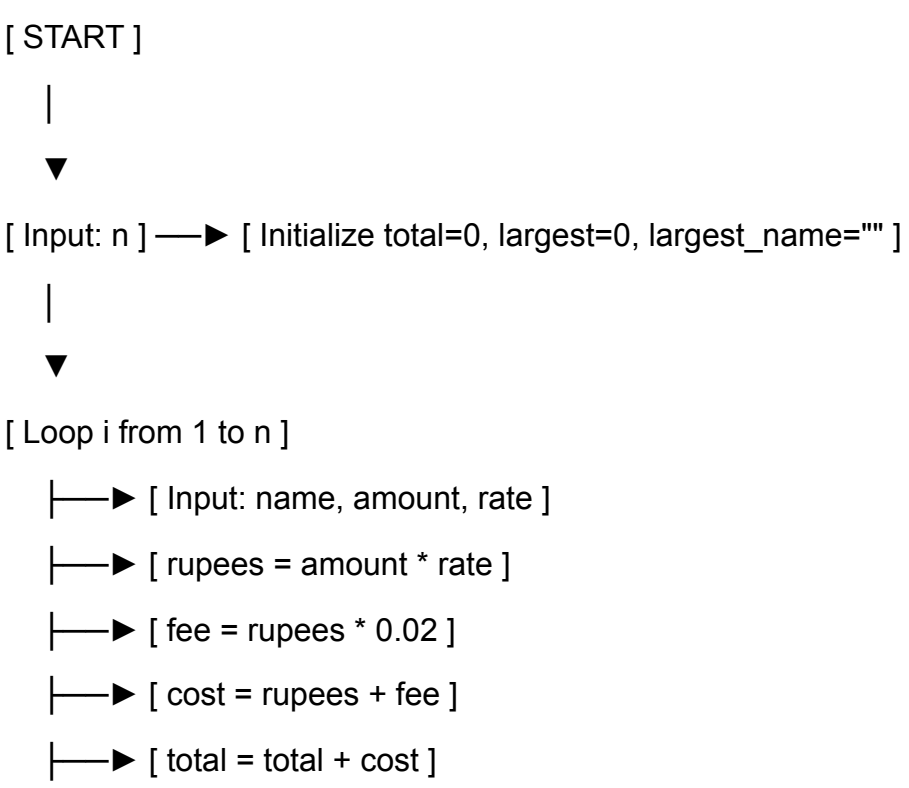
- If total <= budget, print "Within Budget".

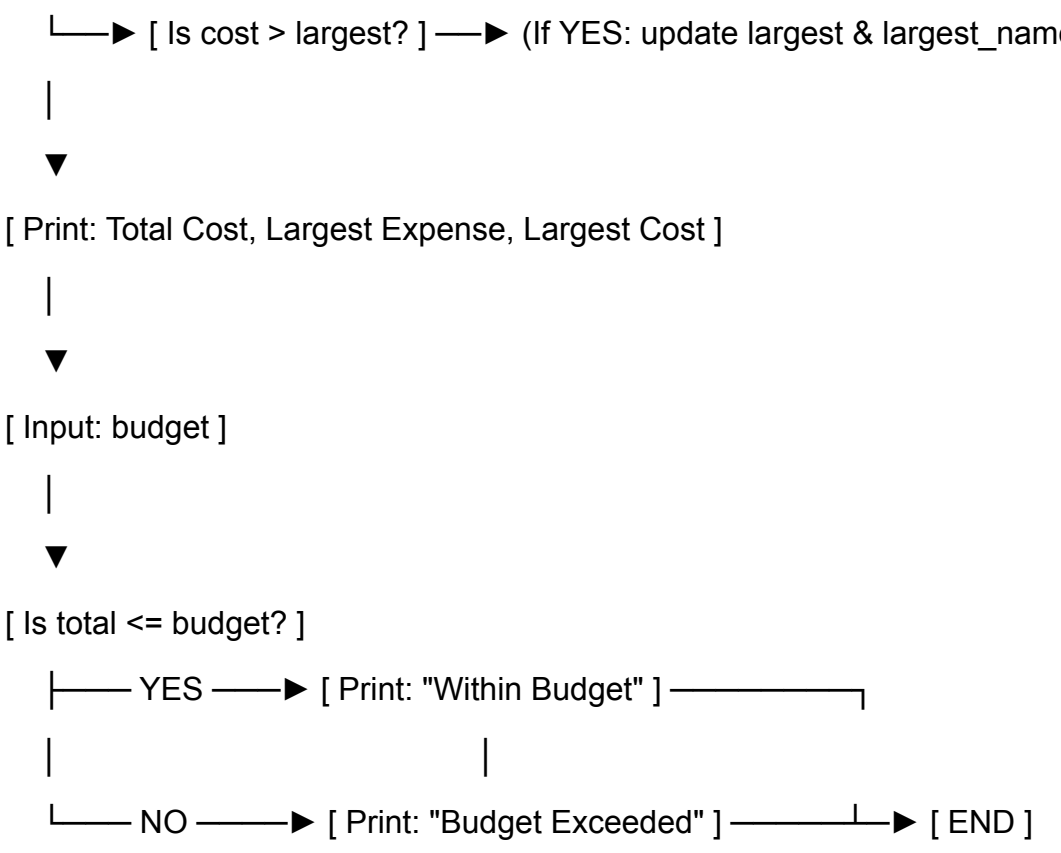
- Else, print "Budget Exceeded".

8. **End** the program.

Flowchart Representation

Plaintext





Pseudocode

Plaintext

```
BEGIN
    PRINT "Enter number of expenses: "
    READ n
    total = 0
```

```
largest = 0
largest_name = ""

FOR i = 1 TO n DO
    PRINT "Enter expense category: "
    READ name
    PRINT "Enter foreign currency amount: "
    READ amount
    PRINT "Enter exchange rate in Rs.: "
    READ rate

    rupees = amount * rate
    fee = rupees * 0.02
    cost = rupees + fee

    total = total + cost

    IF cost > largest THEN
        largest = cost
```

```
        largest_name = name
      ENDIF
    ENDFOR

    PRINT "Total Travel Cost: Rs.", total
    PRINT "Largest Expense:", largest_name
    PRINT "Largest Expense Cost: Rs.", largest

    PRINT "Enter travel budget in Rs.: "
    READ budget

    IF total <= budget THEN
      PRINT "Within Budget"
    ELSE
      PRINT "Budget Exceeded"
    ENDIF
END
```

3. Python Source Code, Test Cases, and Outputs

Python Source Code

Python

```
# Foreign Travel Expense Tracking and Budget Management System

n = int(input("Enter number of expenses: "))
total = 0
largest = 0
largest_name = ""

for i in range(n):
    name = input("Enter expense category: ")
    amount = float(input("Enter foreign currency amount: "))
    rate = float(input("Enter exchange rate in Rs.: "))

    rupees = amount * rate
    fee = rupees * 0.02
    cost = rupees + fee

    total = total + cost
```

```
    if cost > largest:
        largest = cost
        largest_name = name

print("Total Travel Cost: Rs.", total)
print("Largest Expense:", largest_name)
print("Largest Expense Cost: Rs.", largest)

budget = float(input("Enter travel budget in Rs.: "))

if total <= budget:
    print("Within Budget")
else:
    print("Budget Exceeded")
```

Test Cases Table

Test Case ID	Inputs (Category, Amount, Rate)	Total Cost (Rs.)	Largest Expense	Budget (Rs.)	Evaluation Result	Status

TC_01	Hotel: 100 @ 80 Food: 50 @ 80	12240.0	Hotel (Rs. 8160.0)	15000.0	Within Budget	Pass
TC_02	Flight: 500 @ 85 Shopping: 200 @ 85	61200.0	Flight (Rs. 43350.0)	50000.0	Budget Exceeded	Pass
TC_03	Transport: 200 @ 10	2040.0	Transport (Rs. 2040.0)	2000.0	Budget Exceeded	Pass
TC_04	Sightseeing: 150 @ 50	7650.0	Sightseeing (Rs. 7650.0)	10000.0	Within Budget	Pass

Simulated Terminal Output

Bash

Enter number of expenses: 2

Enter expense category: Hotel

Enter foreign currency amount: 100

Enter exchange rate in Rs.: 80

Enter expense category: Food

Enter foreign currency amount: 50

Enter exchange rate in Rs.: 80

Total Travel Cost: Rs. 12240.0

Largest Expense: Hotel

Largest Expense Cost: Rs. 8160.0

Enter travel budget in Rs.: 15000

Within Budget

4. Conclusion

This Python script successfully automates multi-currency financial tracking by combining looping structures, mathematical conversion formulas, percentage fee additions, dynamic maximum-value searching, and budget condition checks. It serves as an efficient, robust prototype for personal finance apps, travel budgeting software, and corporate expense management systems.